

Quantum Mechanics

Wave-Particle Duality

Light: wave nature (diffraction, refraction, interference....)
particle nature (photoelectric effect, Compton effect)
└─ called “photon”

Electron: particle nature ($m_0, q \dots$)
wave nature (diffraction, refraction, interference...)

de Broglie's matter wave: All particles have the properties of wave.

$$\lambda = \frac{h}{mv} \quad \longrightarrow \quad p = \frac{h}{\lambda} \quad \text{de Broglie's hypothesis}$$

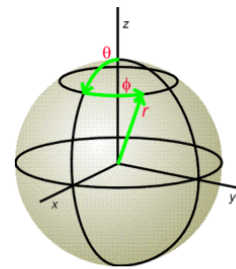
Wave Functions

Wavefunctions

- A particle **trajectory** is a classical concept.
- In Quantum Mechanics, a “particle” (e.g. an electron) does not follow a definite trajectory $\{\mathbf{r}(t), \mathbf{p}(t)\}$, but rather it is best described as being **distributed through space like a wave**.

Definitions

- **Wavefunction** (ψ) – a wave representing the spatial distribution of a “particle”.
- e.g. electrons in an atom are described by a wavefunction centred on the nucleus.
- ψ is a function of the coordinates defining the position of the classical particle:
 - 1-D $\psi(x)$
 - 3-D $\psi(x,y,z) = \psi(\mathbf{r}) = \psi(r, \theta, \phi)$ (e.g. atoms)
- ψ may be time dependent – e.g. $\psi(x,y,z,t)$



The Importance of ψ

- ψ completely defines the system (e.g. electron in an atom or molecule).
- If ψ is known, we can determine any observable property (e.g. energy, vibrational frequencies, ...) of the system.
- QM provides the tools to determine ψ computationally, to interpret ψ and to use ψ to determine properties of the system.

To get full knowledge of a problem **POSITION** and **MOMENTUM** need to be known

For **QUANTUM particles** its Uncertain $\rightarrow \Delta x \Delta p_x \geq \frac{\hbar}{2}$

First Postulate Introduce a parameter called

$$\Psi(x)$$

- Δx and Δp , are not absolute values, they represents the accuracy of measurement.
- The state of a Quantum Object is completely specified by a Wave Function $\psi(x)$

$\Psi^*(x)\Psi(x)$ is the probability density.

For Quantum Particles probability gives full information.

Probability that the particle can be found in the interval dx

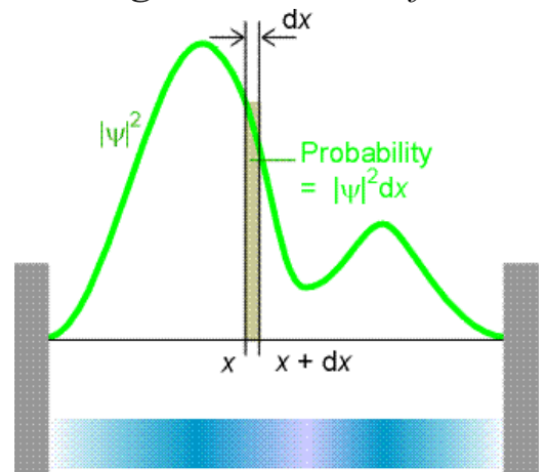
$$\Psi^*(x)\Psi(x)dx$$

A given wavefunction of the form $\Psi(x) = Af(x)$ needs to get *normalized such that total Probability = 1*, using the condition for **NORMALIZATION**

$$\int_{all} \Psi^*(x)\Psi(x)dx = 1$$

Suitable A is obtained

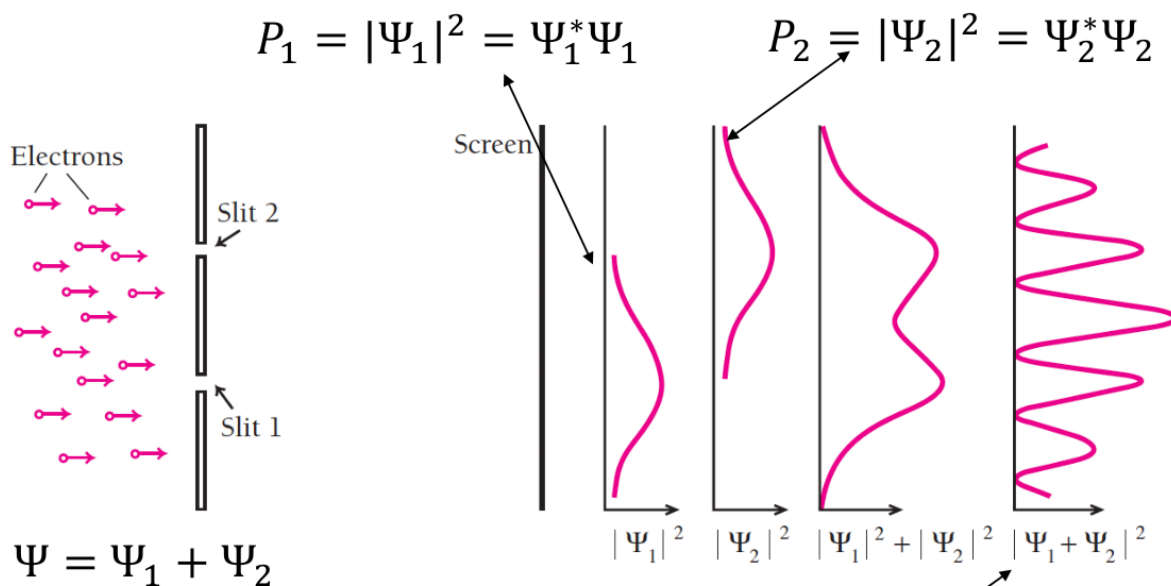
Probability must be real and positive
($0 \leq P \leq 1$).



Linearity and Superposition

Let $\Psi_1(x)$ and $\Psi_2(x)$ represent the two wave functions of a quantum system, then the linear superposition will also be a valid wavefunction for the same quantum system.

$$\Psi(x) = a_1\Psi_1(x) + a_2\Psi_2(x)$$



$$\begin{aligned} P &= |\Psi|^2 = |\Psi_1 + \Psi_2|^2 = (\Psi_1^* + \Psi_2^*)(\Psi_1 + \Psi_2) \\ &= \Psi_1^*\Psi_1 + \Psi_2^*\Psi_2 + \Psi_1^*\Psi_2 + \Psi_2^*\Psi_1 \\ &= P_1 + P_2 + \Psi_1^*\Psi_2 + \Psi_2^*\Psi_1 \end{aligned}$$

The wave functions Ψ_1 and Ψ_2 add to produce the intensity at the screen, not the probability densities $|\Psi_1|^2$ and $|\Psi_2|^2$.

Expectation value

Let $\Psi(x)$ describe the wavefunction of a quantum system, average value of the position (x) is given by the expectation value integral:

$$\langle x \rangle = \int \Psi^*(x) x \Psi(x) dx$$

Expectation value of position of a quantum particle

Probability that the particle can be found in the interval dx

$$\Psi^*(x) \Psi(x) dx$$

What is the expectation value of the position of the particle ?

$$\langle x \rangle = \frac{P_1 x_1 + P_2 x_2}{P_1 + P_2}$$

$$\langle x \rangle = \frac{\int_{all} x \Psi^*(x) \Psi(x) dx}{\int_{all} \Psi^*(x) \Psi(x) dx}$$

One-electron Schrödinger equation

General Formulation

- (1) There exist a wavefunction, $\Psi(x, y, z, t)$, describing the dynamic behavior of the system. $\swarrow$ mathematically complex quantity
- (2) Ψ and $\nabla\Psi$ must be continuous, finite, and single-valued for all values of x, y, z , and t .
- (3) $\Psi^* \Psi dV = |\Psi|^2 dV$: the probability that the particle will be found in the spatial volume element dV .
- $$\int \Psi^* \Psi dV = 1: \text{ normalization}$$

Operators

Operators and Observables

- If ψ is the wavefunction representing a system, we can write:

$$\hat{Q}\psi = Q\psi$$

where Q – “observable” property of system (e.g. energy, momentum, dipole moment ...)

$\hat{Q}$ – operator corresponding to observable Q .

- This is an **eigenvalue equation** and can be rewritten as:

$$\hat{Q}(\psi) = Q \times \psi$$

operator Q acting on function ψ multiplied
function ψ by a number Q
(eigenfunction) (eigenvalue)

(Note: ψ can't be cancelled).

Examples: $d/dx (e^{ax}) = a e^{ax}$

Quantum Mechanical Position and Momentum Operators

- Operator for position in the x-direction is just multiplication by x

$$\hat{x}\psi = x\psi$$

- Operator for linear momentum in the x-direction: $\hat{p}_x = \left(\frac{\hbar}{i}\right) \left(\frac{d}{dx}\right)$

$$\hat{p}_x\psi = p_x\psi \Rightarrow \frac{\hbar}{i} \cdot \frac{d\psi}{dx} = p_x\psi$$

(solve first order differential equation $\Rightarrow \psi, p_x$).

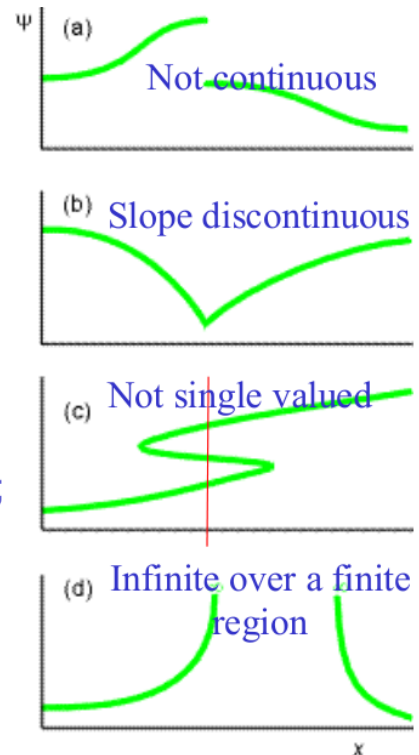
Physical properties		Operators	
Name of Operator	Observables	Operators	Symbols
Position	Position with x coordinate	x	$\hat{x}$
Momentum	x component of momentum	$-i\hbar \frac{\partial}{\partial x}$	$\widehat{p_x}$
Kinetic Energy	Kinetic Energy	$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$	$\hat{T}$
Potential Energy	Potential Energy	V(x)	$\hat{V}$
Total Energy (E)	Hamiltonian Operator (Time-independent)	$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$	$\hat{H}$
Total Energy (E)	Hamiltonian Operator (Time-dependent)	$-i\hbar \frac{\partial}{\partial t}$	$\hat{H}$

The Wave Function

The Wavefunction

The Born interpretation of ψ places restrictions on the form of the wavefunction:

- (a) ψ must be continuous (no breaks);
- (b) The gradient of ψ ($d\psi/dx$) must be continuous (no kinks);
- (c) ψ must have a single value at any point in space;
- (d) ψ must be finite everywhere;
- (e) ψ cannot be zero everywhere.



- Other restrictions (**boundary conditions**) depend on the exact system.
- These restrictions on ψ mean that only certain wavefunctions and $\therefore$ only certain energies of the system are allowed.

$\therefore$ Quantization of $\psi \Rightarrow$ Quantization of E

To find ψ and calculate the properties (observables) of a system:

1. Construct relevant operator $\hat{Q}$
2. Set up equation $\hat{Q}\psi = Q\psi$
3. Solve equation $\Rightarrow$ allowed values of ψ and Q .

Schrodinger Equation

The Schrödinger Equation (1926)

- The central equation in Quantum Mechanics.
- Observable = total energy of system.

Schrödinger Equation $\hat{H}\psi = E\psi$ $\hat{H}$ Hamiltonian Operator
E Total Energy

where $\hat{H} = \hat{T} + \hat{V}$ and $E = T + V$.

- SE can be set up for any physical system.
- The form of $\hat{H}$ depends on the system.
- Solve SE $\Rightarrow \psi$ and corresponding E.

Examples

1. Particle Moving in 1-D $\psi(x)$

$$\hat{H}\psi = \hat{T}\psi + \hat{V}\psi = E\psi \longrightarrow -\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}\right) + V(x)\psi = E\psi$$

- The form of $V(x)$ depends on the physical situation:
 - Finite well $V(x) = 0$ for all a range of x and finite for rest.
 - Infinite well $V(x) = 0$ for all a range of x and infinite for rest.
 - Harmonic oscillator $V(x) = \frac{1}{2}kx^2$

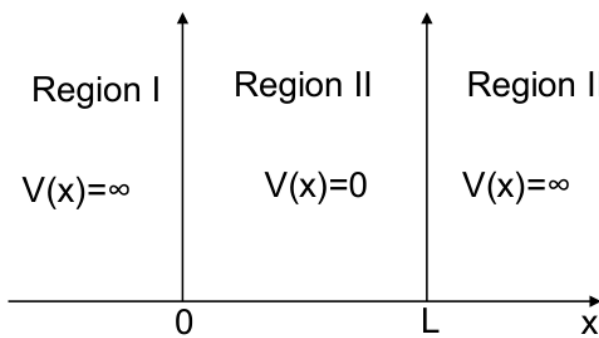
The Particle in a Box

Particle in 1-dimensional box

Time Independent Schrödinger Equation

$$\underbrace{\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2}}_{\text{KE}} + \underbrace{V(x)\psi}_{\text{PE}} = \underbrace{E\psi}_{\text{TE}}$$

- Infinite walls



Region I Region II Region III

$V(x)=\infty$ $V(x)=0$ $V(x)=\infty$

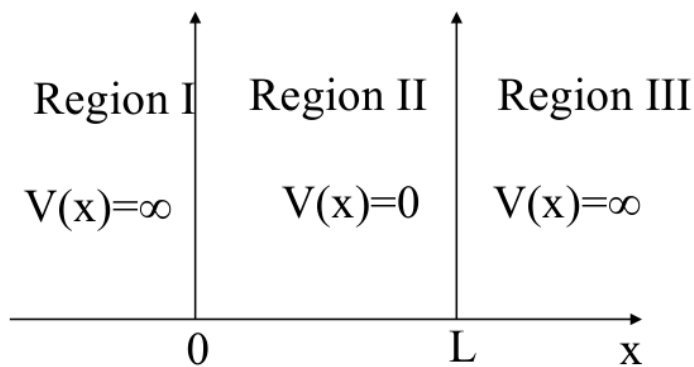
System
Particle of mass m in 1-D box of length L .
Position $x = 0 \rightarrow L$.

Particle cannot escape from box as PE $V(x)=\infty$ for $x = 0, L$ (walls).
PE inside box: $V(x)=0$ for $0 < x < L$.

Restrictions on ψ

- In principle **Schrödinger Eqn.** has an infinite number of solutions.
- So far we have general solutions: any value of $\{\psi, E\}$.
- BUT** – due to the **Born interpretation of ψ** , only certain values of ψ are physically acceptable:
 - outside box ($x < 0, x > L$) $V = \infty \Rightarrow$ impossible for particle to be outside the box
 $\therefore |\psi|^2 = 0 \Rightarrow \psi = 0$ outside box.
 - ψ must be a continuous function
 $\Rightarrow$ **Boundary Conditions** $\psi = 0$ at $x = 0$
 $\psi = 0$ at $x = L$.

Applying boundary conditions: Restrictions on ψ



Region I and III: $\frac{-\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \infty * \psi = E\psi \quad \Rightarrow |\psi|^2 = 0$

Region II:

$$\frac{d^2\Psi(x)}{dx^2} = -\frac{2mE}{\hbar^2} \Psi(x)$$

$$\frac{d^2\Psi(x)}{dx^2} = -k^2 \Psi(x) \quad k^2 = \frac{2mE}{\hbar^2}$$

At $x = L$ $\psi = 0$

$$0 = A \sin kL$$

But $A \neq 0$

$$kL = n\pi$$

This is similar to the general differential equation:

$$\psi = A \sin kx + B \cos kx$$

Thus, wave function:

$$\psi_{II} = A \sin \frac{n\pi x}{L}$$

But what is 'A'?

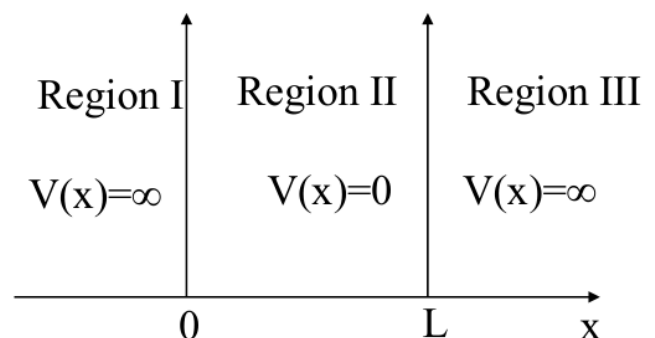
Applying boundary conditions:

At $x = 0 \rightarrow \psi = 0$

$$0 = A \sin 0k + B \cos 0k$$

$$0 = 0 + B * 1$$

$$\Rightarrow B = 0$$



Normalizing wave function:

$$\int_0^L (A \sin kx)^2 dx = 1$$

$$\sin^2 a = \frac{(1 - \cos(2a))}{2}$$

$$\int_0^L A^2 \left(\frac{1}{2} - \frac{\cos 2kx}{2} \right) dx = 1$$

$$|A|^2 \left[\frac{x}{2} - \frac{\sin 2kx}{4k} \right] \Big|_0^L = 1$$

$$|A|^2 \left[\frac{L}{2} - \frac{\sin 2 \frac{n\pi}{L} L}{4 \frac{n\pi}{L}} \right] = 1$$

$$k = \frac{n\pi}{L}$$

$$\sin 2\pi = 0$$

$$|A|^2 \left(\frac{L}{2} \right) = 1$$

$$|A| = \sqrt{\frac{2}{L}}$$

Thus normalized wave function is:

$$\psi_{II} = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

Calculating Energy Levels:

$$k^2 = \frac{2mE}{\hbar^2}$$

$$E = \frac{k^2 \hbar^2}{2m}$$

$$E = \frac{k^2 h^2}{2m4\pi^2}$$

$$k = \frac{n\pi}{L}$$

$$(\hbar = \frac{h}{2\pi})$$

$$E = \frac{n^2 \pi^2}{L^2} \frac{h^2}{2m4\pi^2}$$

Thus Allowed Energies, E_n , are:

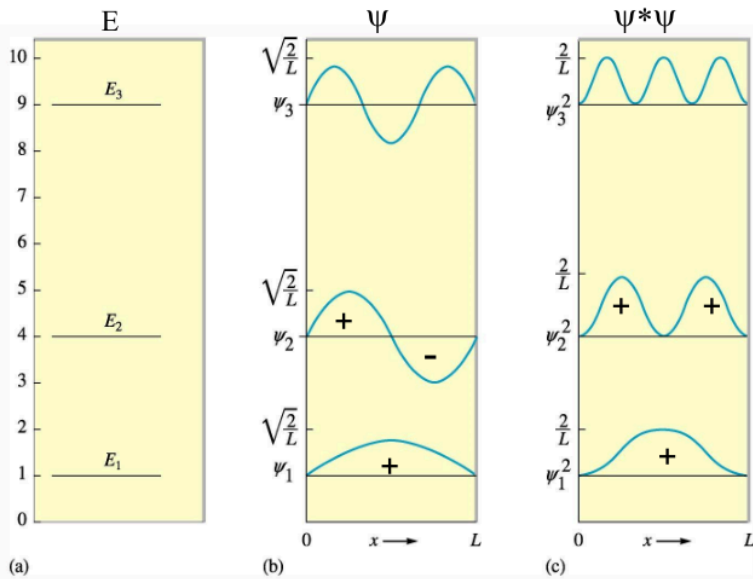
$$E_n = \frac{n^2 h^2}{8mL^2}$$

Particle in a 1-Dimensional Box

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$$\psi_{II} = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

$$|\psi_{II}|^2 = \frac{2}{L} \left(\sin \frac{n\pi x}{L} \right)^2$$



Difference b/w adjacent energy levels:

$$\Delta E = E_{n+1} - E_n = (2n + 1)E_1$$

Expectation Value

Position

$$\psi_n = \sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right)$$

$$\langle x \rangle = \int_0^L x \frac{2}{L} \sin^2 \left(\frac{n\pi x}{L} \right) dx$$

$$= \frac{2}{L} \int_0^L x \left(\frac{1}{2} - \frac{1}{2} \cos \frac{2n\pi x}{L} \right) dx$$

$$= \frac{2}{L} \left[\frac{x^2}{4} - x \frac{\sin \left(\frac{2n\pi x}{L} \right)}{\left(\frac{4n\pi}{L} \right)} - \frac{\cos \left(\frac{2n\pi x}{L} \right)}{8 \left(\frac{n\pi}{L} \right)^2} \right]_0^L$$

$$= \frac{2}{L} \left[\frac{x^2}{4} \right]_0^L - \frac{2}{L} \frac{L^2}{8n^2\pi^2} \left[\frac{2n\pi x}{L} \sin \left(\frac{2n\pi x}{L} \right) + \cos \left(\frac{2n\pi x}{L} \right) \right]_0^L$$

Since $\sin(n\pi) = 0$ and $\cos(n\pi) = 1$, for all the values of n the expectation value of x is

$$\langle x \rangle = \frac{2}{L} \left(\frac{L^2}{4} \right) = \frac{L}{2}$$

Momentum

Momentum

$$\langle p_x \rangle = \int_0^L \left[\left[\frac{2}{L} \right]^{1/2} \sin\left(\frac{n\pi x}{L}\right) \right] \left(-\frac{i\hbar}{2\pi} \frac{d}{dx} \right) \left[\left[\frac{2}{L} \right]^{1/2} \sin\left(\frac{n\pi x}{L}\right) \right] dx$$

$$\langle p_x \rangle = -\frac{2\pi i \hbar}{L^2} \int_0^L \underbrace{\sin\left(\frac{n\pi x}{L}\right)}_{\text{odd}} \underbrace{\cos\left(\frac{n\pi x}{L}\right)}_{\text{even}} dx = 0$$

Also try with

$$\sin\left(\frac{n\pi x}{L}\right) = u$$

$$\frac{n\pi}{L} \cos\left(\frac{n\pi x}{L}\right) dx = du$$

Trigonometric Functions	
Even Functions $f(-x) = f(x)$	Odd Functions $f(-x) = -f(x)$
$\cos(-x) = \cos x$ $\sec(-x) = \sec x$	$\sin(-x) = -\sin x$ $\csc(-x) = -\csc x$ $\tan(-x) = -\tan x$ $\cot(-x) = -\cot x$

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2\pi} \sin^2\left(\frac{n\pi x}{L}\right) = 0$$

Finite Potential Well

Finite Potential Well

Schrödinger Equation $\rightarrow$
$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + U(x)\psi(x) = E\psi(x)$$

I. $x < 0; U(x) = U$

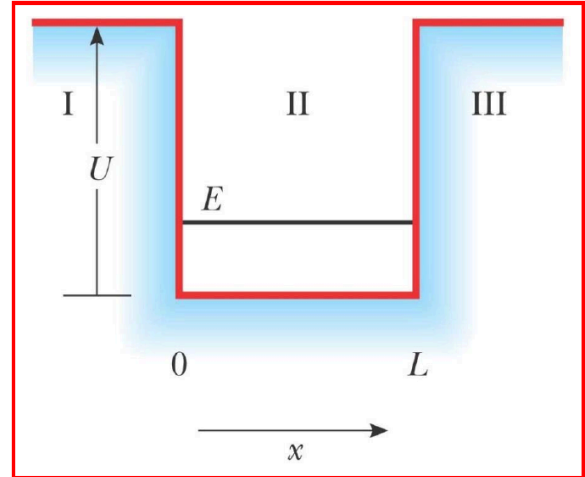
$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_I}{dx^2} + U\psi_I = E\psi_I$$

I. $0 < x < L; U(x) = 0$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_{II}}{dx^2} = E\psi_{II}$$

III. $x > L; U(x) = U$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_{III}}{dx^2} + U\psi_{III} = E\psi_{III}$$



We also assume that energy of the particle, E , is less than the “height” of the barrier, i.e.

$$E < U$$

Finite Potential Well: Regions I and III

- The Schrödinger equation for these regions is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U\psi = E\psi$$

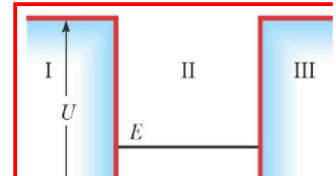
- It can be re-written as

$$\frac{d^2\psi}{dx^2} = \frac{2m(U - E)}{\hbar^2} \psi = C^2\psi, \quad \text{where } C^2 \equiv \frac{2m(U - E)}{\hbar^2}$$

- The general solution of this equation is

$$\psi(x) = Ae^{Cx} + Be^{-Cx}$$

– A and B are constants



- Requiring that wavefunction was finite at $x \rightarrow \infty$ and $x \rightarrow -\infty$, we can show that
- In region I, $B = 0$, and $\psi_I(x) = Ae^{Cx}$
 - This is necessary to avoid an infinite value for $\psi(x)$ for large negative values of x
- In region III, $A = 0$, and $\psi_{III}(x) = Be^{-Cx}$
 - This is necessary to avoid an infinite value for $\psi(x)$ for large positive values of x

The wavefunction and its derivative must be *single-valued* for all x

- At $x = 0$ and $x = L$ the wave function and its derivative have to be *continuous*

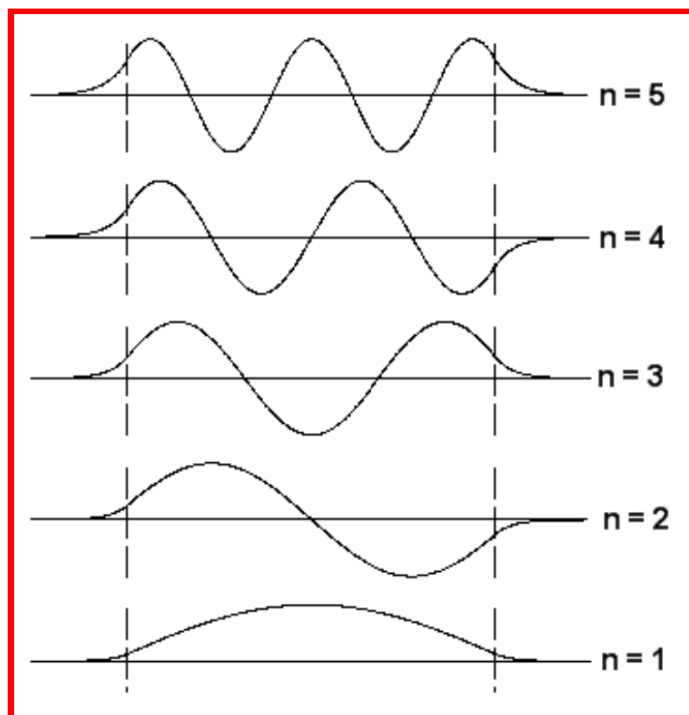
Thus, we have to equate “parts” of the wavefunction and its derivative at $x = 0, L$

- This, together with normalization condition, allows to determine the constants and the equation for energy of the particle

$$\begin{aligned}\psi_I(0) &= \psi_{II}(0) \\ \frac{d\psi_I}{dx}(0) &= \frac{d\psi_{II}}{dx}(0) \\ \psi_{II}(L) &= \psi_{III}(L) \\ \frac{d\psi_{II}}{dx}(L) &= \frac{d\psi_{III}}{dx}(L)\end{aligned}$$

$$\psi_{II}(x) = F \sin kx + G \cos kx$$

$$\psi_I(x) = Ae^{Cx}$$



$$\psi_{III}(x) = Be^{-Cx}$$

Graphical Results for $\psi(x)$

The particle penetrates the walls!

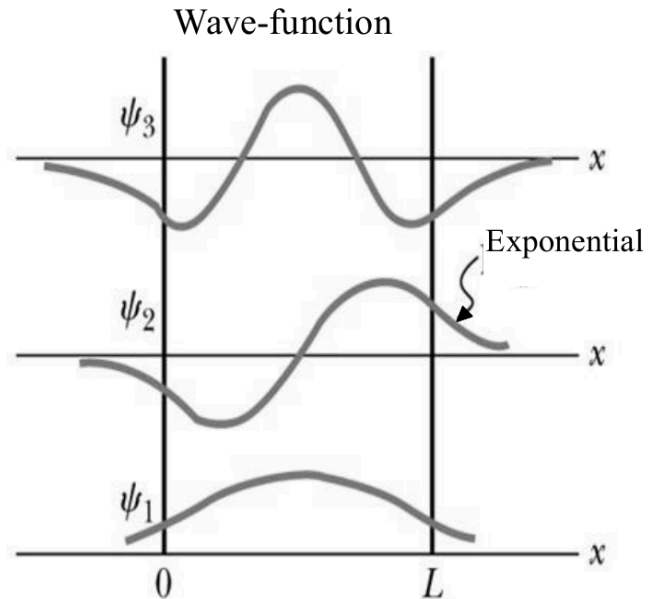
This violates classical physics!

The penetration depth is the distance outside the potential well where the probability decreases to about $1/e$. It's given by:

$$\delta x \approx \frac{1}{C} = \frac{\hbar}{\sqrt{2m(U - E)}}$$

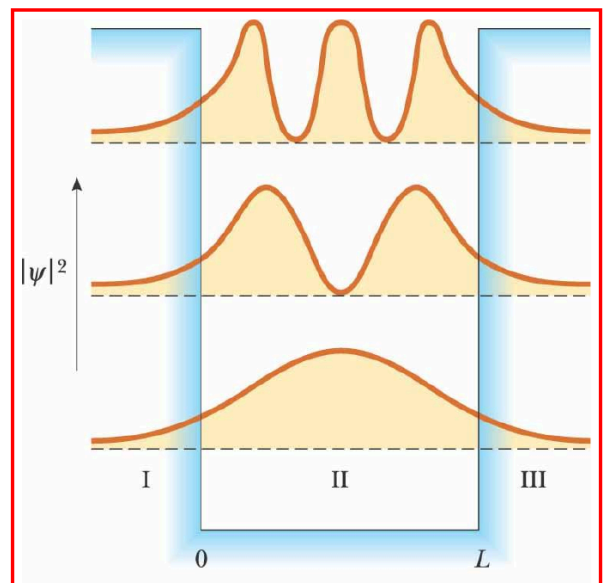
Note that the penetration distance is proportional to Planck's constant.
 $\hbar = 10^{-34} \text{ Joule} - \text{seconds} \rightarrow$ usually an extremely tiny distance.

As $n \uparrow \rightarrow E \uparrow \rightarrow$ Penetration depth increases more and more.

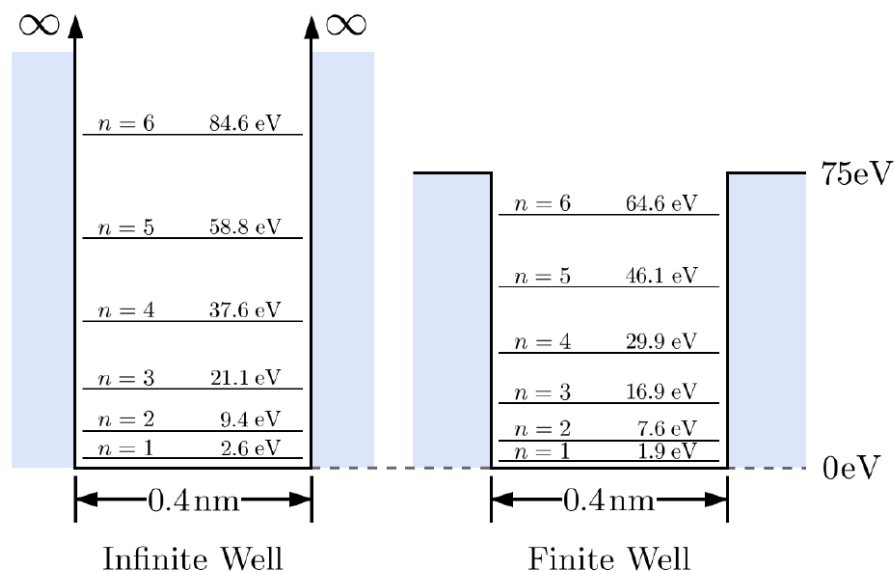


$$|\psi(x)|^2$$

- The probability densities for the lowest three states are shown
- The functions are smooth at the boundaries
- Outside the box, the probability to find the particle decreases exponentially, **but it is not zero!**



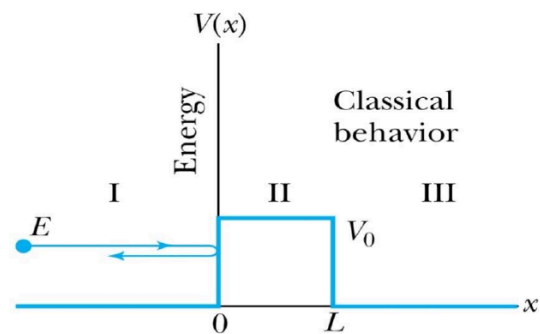
Comparison with infinite-well potential: The energy of state n is lower in the finite square well potential of the same width.



Tunneling

Consider a particle of energy E approaching a potential barrier of height V_0 , and the potential everywhere else is zero.

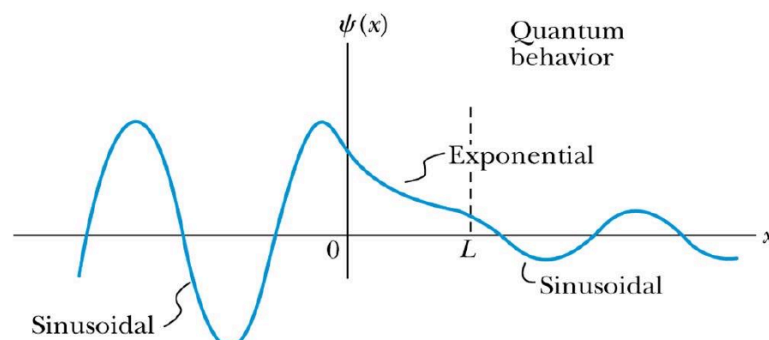
Also $E < V_0$.



The solutions in Regions I and III are sinusoids.

The wave function in region II becomes: $\psi_{II} = C e^{\alpha x} + D e^{-\alpha x}$

Enforcing continuity at the boundaries:



Tunneling is key in radioactivity.

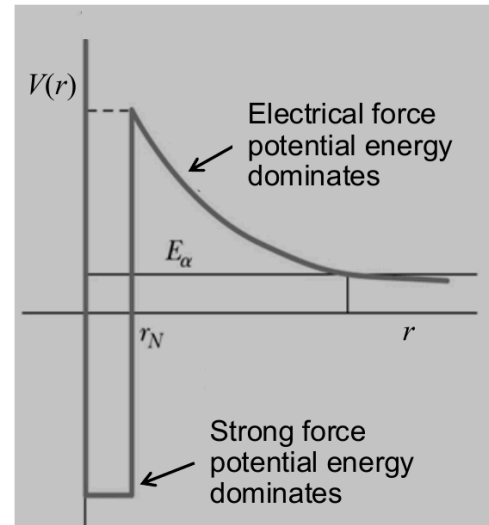
Radioactive Decay

Inside a nucleus, there's only positive charge, which is repulsive.

But inside the nucleus, the strong, short-range attractive nuclear force (~a square well) dominates the repulsive electrical force.

The potential barrier at the nuclear radius is several times greater than the energy of an alpha particle, E_α .

In quantum mechanics, however, the alpha particle can **tunnel** through the barrier.



α -particle potential energy curve

This is an excellent model for radioactive decay!

Harmonic Oscillator

Simple Harmonic Motion from the View of Quantum Mechanics

In order to know the wave function that describe the harmonic vibrational motion for the simple harmonic oscillator and the allowed energy levels, we have to solve the Schrödinger equation

$$\hat{H}\psi = E\psi$$

the Hamiltonian operator is given by

$$\hat{H} = \hat{T} + \hat{V}$$

the kinetic energy operator is given by

$$\hat{T} = -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2}$$

notice that, we are assuming that the motion of the simple harmonic oscillator (e.g. the vibration of a diatomic molecule) is in one direction and we assumed that the motion is along the x - axis.

The potential energy operator

$$\hat{V} = \frac{1}{2} kx^2$$

Now substituting $\hat{T}$ and $\hat{V}$ into the one-dimensional Schrödinger equation gives $-\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} \psi(x) = E \psi(x)$

After some effort, the eigenfunctions are $n = 0, 1, 2, \dots$

$$\psi_n(x) = N_n H_n\left(\frac{\alpha}{2} x\right) e^{-\frac{\alpha x^2}{2}}$$

where H_n is the Hermite polynomial of degree n .

α is defined as:

$$\alpha = \frac{2\pi\mu\nu}{\hbar}$$

N_n is the normalization constant which is defined as:

$$N_n = (2^n n!)^{-\frac{1}{2}} \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}$$

Table 5.2 Some Hermite Polynomials

n	$H_n(y)$	α_n	E_n
0	1	1	$\frac{1}{2}\hbar\nu$
1	$2y$	3	$\frac{3}{2}\hbar\nu$
2	$4y^2 - 2$	5	$\frac{5}{2}\hbar\nu$
3	$8y^3 - 12y$	7	$\frac{7}{2}\hbar\nu$
4	$16y^4 - 48y^2 + 12$	9	$\frac{9}{2}\hbar\nu$
5	$32y^5 - 160y^3 + 120y$	11	$\frac{11}{2}\hbar\nu$

$$y = \sqrt{\alpha} x$$

The eigenvalues are

$$E_n = \left(n + \frac{1}{2}\right) \hbar\nu \quad n = 0, 1, 2, 3, \dots$$

where n = the vibrational quantum number.

IMPORTANT : The energy is restricted to the quantized values

Recall that,

$$\nu = \frac{\omega}{2\pi} \quad \text{and} \quad \omega = \sqrt{k/\mu} \Rightarrow \nu = \frac{1}{2\pi} \sqrt{k/\mu}$$

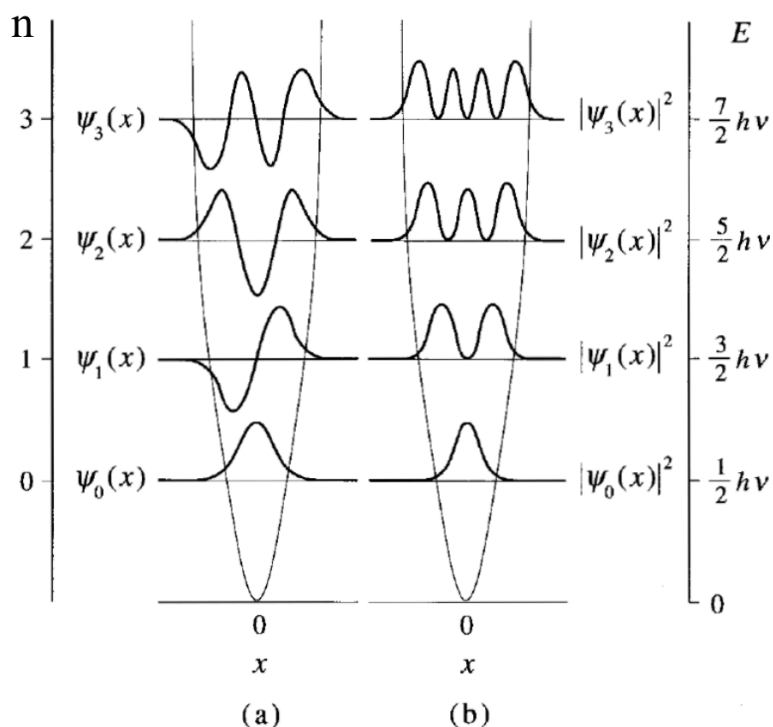
And

$$\hbar = \frac{h}{2\pi} \Rightarrow h = \hbar 2\pi$$

Therefore,

$$E_v = \hbar(k/\mu)^{\frac{1}{2}} \left(n + \frac{1}{2}\right)$$

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right)$$

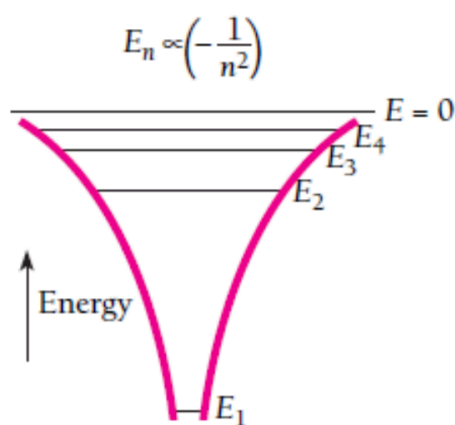


Levels are **equally spaced**:

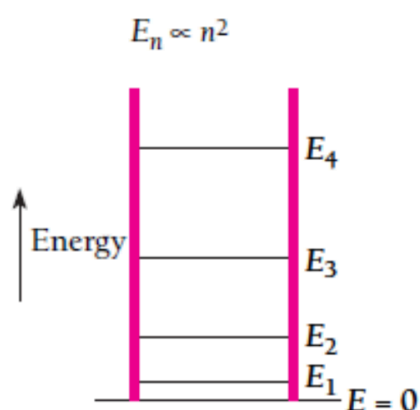
$$E_{n+1} - E_n = \hbar\omega = h\nu$$

(a) The normalized harmonic-oscillator wave functions.

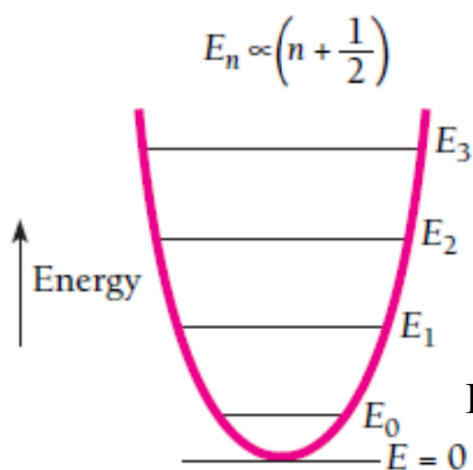
(b) The probability densities for a harmonic oscillator.



Coulombic Potential



Square well Potential



Harmonic Potential